

Product Manual

Bally / Williams

P/N TPP-1042

Trough Opto

Transmitter and Receiver Boards



Tangles'
Pin Parts

TPP-1042-TX Replaces A-18617-1 and A-18617

TPP-1042-RX Replaces A-18618-1 and A-18618



The Tangles' Trough Optos are a drop in replacement part compatible with A-18618, A18618-1, A-18617 and A-18617-1.

The boards are sold individually as Transmitter (TX), Receiver (RX) and in 4 ball (5 opto) and 6 ball (7 opto) variations.

See the compatibility guide below to determine if your machine requires the 4 ball or 6 ball option.

These opto boards are common on late model Bally Williams machines.

The LEDs and phototransistors used in these boards are 880nm compatible making the new boards compatible with the original Bally Williams boards. This means a new Tangles TX board can be paired with an original RX board for example.

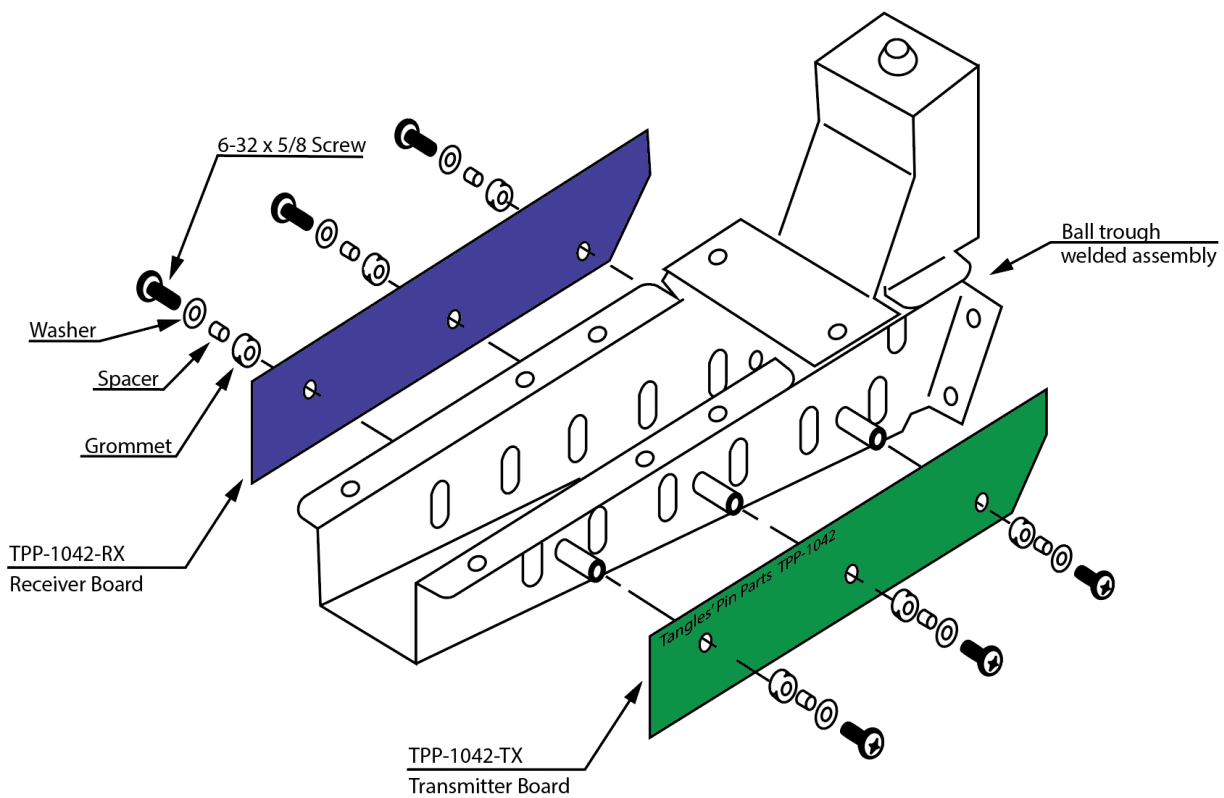
The Tangles boards are optionally sold with mounting hardware, but the original mounting hardware can be used on the new boards.

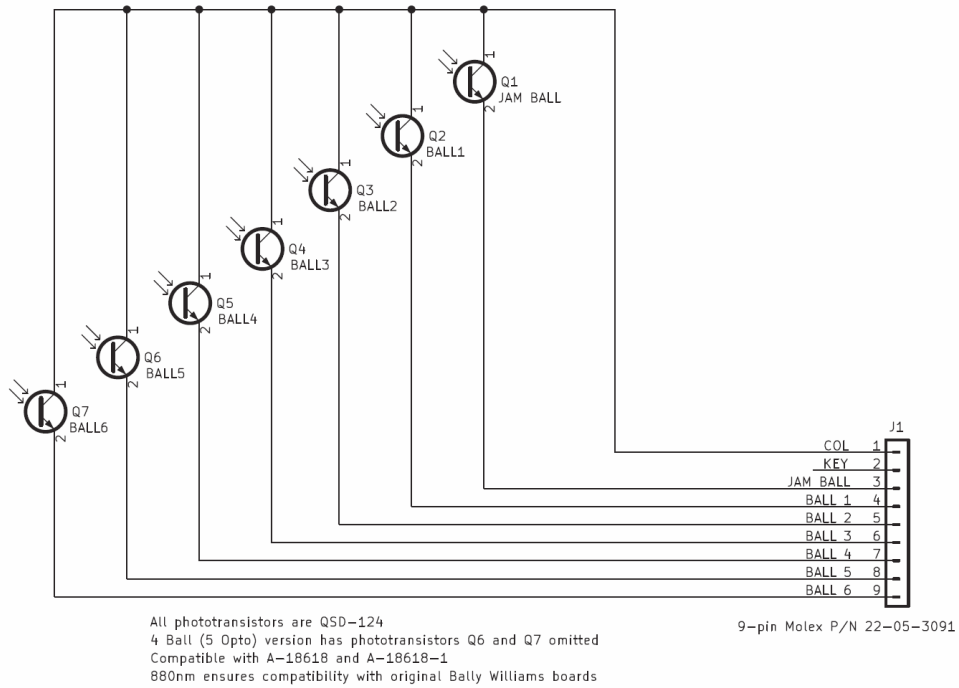
INSTALLATION

Installation of the boards is best achieved by removing the ball trough from the machine. The transmitter board (green) is located on the front of the metal trough (nearest the coin door). The receiver board (blue) is located on the rear of the metal trough.

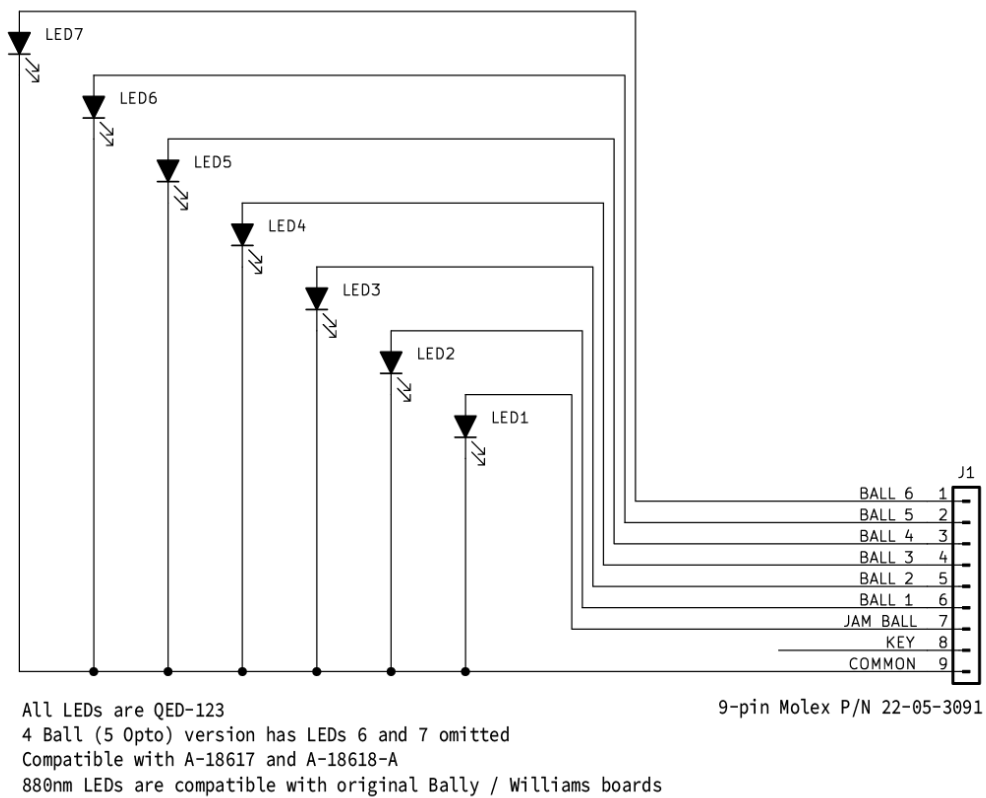
- Ensure power is off
- Remove all balls
- Unscrew and remove the apron

- Unplug the two 9 way Molex connectors from the opto receiver and transmitter boards.
- Remove the 6 machine thread screws from the top side of the playfield
- Remove the two self tapping screws from the underside of the playfield.
- The ball trough and opto boards will come free of the playfield (but still be bound by the wires to the ball eject coil.
- Remove the three #6 screws holding the opto board to the metal trough (TX or RX).
- Install the three rubber grommets into the new opto board.
- Insert three metal spacers into the rubber grommets.
- Screw the new opto board into place using three #6 x 5/8" screws.
- Repeat for the other board (TX or RX)
- Note the RX board (blue) is installed at the back of the metal trough.
- Note the TX board (green) is installed on the front of the metal trough
- Reinstall the metal trough into the playfield (reverse of the removal).
- Re-install the apron.
- Power the machine on and go into switch test mode.
- Install the balls one at a time and note the activation of each of the trough ball switches (optos).





TPP-1042-RX Schematic



TPP-1042-TX Schematic

Compatibility

4 Ball (5 Opto)

Attack From Mars,
Cactus Canyon,
Champion Pub,
Cirque Voltaire,
Congo,
Dirty Harry,
I500,
Jackbot,
Johnny Mnemonic,
Junk Yard,
Monster Bash,
Medieval Madness,
NBA Fastbreak,
No Fear,
Revenge From Mars,
Safe Cracker,
Scared Stiff,
Star Wars Episode 1,
Theatre of Magic,
Tales of the Arabian Nights,
Who Dunnit

6 Ball (7 Opto)

Corvette,
Flintstones,
No Good Gofers,
Road Show,
The Shadow,
World Cup Soccer